The Glass Ceiling: Does Education Equitably Predict Wages for Ontario Immigrants?

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Contents

0.1 Project overview

This project examines whether **education** predicts **wages** in a similar way for **immigrants in Ontario** as it might for other groups—or whether systematic differences suggest barriers consistent with a **glass ceiling** (e.g., credentials not translating equally into earnings). We combine exploratory work, inference, and modeling to connect Ontario labour data to that question.

0.2 Core team

Role	Name	Report segment (draft)
Student 1	Moiz Mohammad	Foundation and visuals (pp. 1–2.5)
Student 2	Asad Mirza	Inference (pp. 2.5–5)
Student 3	Ishan Amin	Model, validation, conclusion (pp. 5–7.5+)

0.3 Research questions

1. How strongly is educational attainment associated with wages among Ontario residents with an immigrant background?
2. Does the education–wage gradient differ by gender, region, immigrant class, or years since landing?
3. After controlling for relevant covariates, do disparities remain that suggest education does **not** equitably predict wages for this population?

LABEL — Page budget (remove this block before submission): Each person should target roughly **2.5 pages** of *content* (text + code + plots). In practice that is often about **one page of prose** and ~**1.5 pages of output/figures**. If the PDF runs short, prefer **larger figures** (chunk options `fig.width` / `fig.height`), **policy context** (e.g., minimum wage, immigration targets), and **interpretation** (what p-values and coefficients mean for a student in Ontario), not filler.

```
# Shared packages; add others (e.g., boot, broom, modelsummary) as needed.
library(tidyverse)
library(knitr)
library(kableExtra)  # optional: nicer tables in PDF
```

```
# MOIZ: readr::read_csv() is preferred over read.csv(): faster, tibbles, consistent parsing, p
# trim_ws = TRUE helps columns like WAGE RATE / EDUCATION that have leading spaces in this fil
# show_col_types = FALSE keeps knit/PDF output shorter (omit if you want to see column guesses
wages_raw <- readr::read_csv(
  "data.csv",
  trim_ws = TRUE,
  show_col_types = FALSE
)

wages <- wages_raw
```

0.4 LABEL: Student 1 (Moiz) — The Foundation and Visuals

0.4.1 Target: approximately pages 1–2.5 (background, dictionary, EDA)

0.4.2 Background (Ontario Ministry of Labour)

LABEL (remove before submission): Aim for ~0.5 page. Summarize why the Ontario Ministry of Labour tracks wages by education, who the intended users are (policy, employers, workers, researchers), and how that motivates your variables. Cite the Ministry pages or publications you use.

(Replace this paragraph with Moiz's narrative.)

0.4.3 Data dictionary

LABEL (remove before submission): ~0.5 page. Define each analysis variable (e.g., `wage_rate`, `education_level`, year, geography). State how you treated missing values, duplicates, and aggregate **"Total"** rows (dropped vs. kept).

Variable name (in R)	Description	Type / units	Notes (missing, "Total" rows, etc.)
`wage_rate`			
`education_level`			
(add rows)			

0.4.4 Exploratory data analysis

LABEL (remove before submission): ~1.5 pages including Table 1 and two graphs. Use chunk options to resize figures if needed (defaults are already 8 x 5 in `setup`).

```
# MOIZ: Table 1 --- summary statistics (mean, median, SD) of wages by education level.
# wages %>%
#   group_by(education_level) %>%
#   summarise(
#     n = n(),
#     mean_wage = mean(wage_rate, na.rm = TRUE),
#     median_wage = median(wage_rate, na.rm = TRUE),
#     sd_wage = sd(wage_rate, na.rm = TRUE)
#   ) %>%
#   knitr::kable(digits = 2, caption = "Table 1: Wage summary statistics by education level")
```

```
# MOIZ: Graph 1 --- boxplot of wage by education level (reorder factor levels logically if nee
# ggplot(wages, aes(x = education_level, y = wage_rate)) +
#   geom_boxplot(outlier.alpha = 0.4) +
#   labs(x = "Education level", y = "Wage (specify: hourly / weekly)") +
#   theme_minimal() +
#   theme(axis.text.x = element_text(angle = 35, hjust = 1))
```

```
# MOIZ: Graph 2 --- line graph: mean wage by year for "University degree" vs "High school" (ma
# wages_trend <- wages %>%
#   filter(education_level %in% c("University degree", "High school")) %>%
#   group_by(year, education_level) %>%
#   summarise(mean_wage = mean(wage_rate, na.rm = TRUE), .groups = "drop")
# ggplot(wages_trend, aes(x = year, y = mean_wage, color = education_level)) +
#   geom_line(linewidth = 1) +
#   geom_point() +
#   labs(x = "Year", y = "Mean wage", color = "Education") +
#   theme_minimal()
```

0.5 LABEL: Student 2 (Asad) — The Inference Specialist

0.5.1 Target: approximately pages 2.5–5 (tests, CIs, bootstrap)

0.5.2 Hypothesis testing

We are testing the hypothesis that the mean wage is different for different education levels. Null hypothesis: The mean wage is the same for all education levels. Alternative hypothesis: The mean wage is different for at least one education level.

```
wages_without_total <- wages %>% filter(EDUCATION != "Total, all education levels")

# Summaries use the numeric wage columns (WAGE RATE is only the label "Median hourly wage").
wages_by_education <- wages_without_total %>%
  group_by(EDUCATION) %>%
  summarise(mean_wage = mean(`Both sexes`, na.rm = TRUE), .groups = "drop")
kable(wages_by_education, caption = "Mean of `Both sexes` (median hourly wage) by education le
```

Table 1: Mean of `Both sexes` (median hourly wage) by education level

EDUCATION	mean_wage
0 - 8 years	11.82724
Above bachelor's degree	29.44443
Bachelor's degree	23.83239
High school graduate	16.09474
Post-secondary certificate or diploma	19.40816
Some high school	13.26123
Some post-secondary	15.46509
University degree	25.86886

```
# t-test needs exactly TWO groups on the right-hand side - filter to two education levels, the
two_edu <- wages_without_total %>%
  filter(EDUCATION %in% c("University degree", "High school graduate"))
t.test(`Both sexes` ~ EDUCATION, data = two_edu)
```

```
##
##  Welch Two Sample t-test
##
## data:  Both sexes by EDUCATION
## t = -76.201, df = 8074.1, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means between group High school graduate and grou
## 95 percent confidence interval:
##  -10.025561  -9.522684
## sample estimates:
## mean in group High school graduate    mean in group University degree
##                           16.09474                           25.86886
```

```r
# For ALL education levels at once, use ANOVA (not a single t.test).
aov(`Both sexes` ~ EDUCATION, data = wages_without_total) |> summary()
```

```
##                Df  Sum Sq Mean Sq F value Pr(>F)    
## EDUCATION       7 1294230  184890    3869 <2e-16 ***
## Residuals   36472 1743075      48                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

0.5.3 Confidence intervals

LABEL (remove before submission): ~0.5 page. **95% confidence intervals** for the mean wage in the **highest** and **lowest** education groups (define which categories those are). Explain what "95% confident" means **without** saying "the probability the true mean is in the interval is 95%" in a misleading way.

```r
# ASAD: 95% CIs for mean wage in two education groups (example pattern).
# high <- wages %>% filter(education_level == "University degree")
# low  <- wages %>% filter(education_level == "High school")
# t.test(high$wage_rate)$conf.int
# t.test(low$wage_rate)$conf.int
```

0.5.4 Bootstrapping

LABEL (remove before submission): ~1 page. Use the `boot` package with **B = 1000** (or similar) resamples. Plot a **histogram** of the bootstrap distribution (e.g., bootstrap mean difference or mean wage). Explain why bootstrapping helps when wages may be **skewed** or **non-normal**.

```r
# ASAD: Define a statistic function for boot::boot(); R = 1000.
# library(boot)
# stat_mean <- function(x, i) mean(x[i], na.rm = TRUE)
# boot_out <- boot::boot(wages$wage_rate, statistic = stat_mean, R = 1000)
# hist(boot_out$t, main = "Bootstrap distribution of mean wage", xlab = "Replicate mean")
# boot.ci(boot_out, type = c("perc", "bca"))
```

0.6 LABEL: Student 3 (Ishan) — The Model and Conclusion

0.6.1 Target: approximately pages 5–7.5+ (regression, CV, summary, appendix)

0.6.2 Regression analysis

LABEL (remove before submission): ~1.5 pages. Fit **multiple linear regression**, e.g. `lm(wage ~ education + immigration_status)` (add controls if your dictionary includes them). Present a **regression table**. **Interpret coefficients**: e.g., if a level "University degree" has a coefficient of 500, state that it is associated with **$500 higher weekly wages holding immigration status constant** (only if that matches your outcome scale—adjust wording to your units).

```
# ISHAN: Use factors with a clear reference level; check linear model assumptions in text or b
# fit <- lm(wage_rate ~ education_level + immigration_status, data = wages)
# summary(fit)
# Table: consider broom::tidy(fit), stargazer, gtsummary::tbl_regression, or modelsummary::mod
```

0.6.3 Cross-validation

LABEL (remove before submission): ~0.5 page. **Train/test split**; report **RMSE** (or RMSE on a held-out set) so readers see **prediction error**, not only in-sample fit.

```
# ISHAN: Example: 80/20 split; RMSE on test set.
# set.seed(123)
# n <- nrow(wages)
# train_idx <- sample.int(n, size = floor(0.8 * n))
# train <- wages[train_idx, ]
# test  <- wages[-train_idx, ]
# fit_train <- lm(wage_rate ~ education_level + immigration_status, data = train)
# pred <- predict(fit_train, newdata = test)
# sqrt(mean((test$wage_rate - pred)^2, na.rm = TRUE))
```

0.6.4 Summary and conclusion

LABEL (remove before submission): ~0.5 page. Tie Moiz's EDA, Asad's inference, and your model together: **does Ontario's data suggest a fair return to education**, especially for immigrants? Link back to the **glass ceiling** framing and policy relevance.

(Replace with Ishan's synthesis.)

0.7 Appendix: All code

LABEL (remove before submission): Ensure this lists the final, clean versions of everyone's chunks (knitr will pull in the labeled chunks below). If you rename chunks, update `ref.label` accordingly.

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```

```r
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# ASAD: 95% CIs for mean wage in two education groups (example pattern).
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```

```r
# pred <- predict(fit_train, newdata = test)
# sqrt(mean((test$wage_rate - pred)^2, na.rm = TRUE))
```

0.8 References

- Ontario Ministry of Labour / Government of Ontario. *(Add exact titles and URLs.)*
- Statistics Canada or other data sources. *(Add citations.)*
- Methods: `boot` package (Canty and Ripley); regression texts as assigned.